

Foundation (Jalapeno)

- Q1.** What is the slope of a line perpendicular to $y = 2x + 1$?
(A) $-\frac{1}{2}$
(B) $\frac{1}{2}$
(C) 2
(D) -2
(E) 0
- Q2.** If two lines have slopes of 3 and $-\frac{1}{3}$, what is their relationship?
(A) Parallel
(B) Perpendicular
(C) Skew
(D) Intersecting
(E) Coincident
- Q3.** What is the equation of a line perpendicular to $y = x - 2$ that passes through $(0, 0)$?
(A) $y = -x$
(B) $y = x + 1$
(C) $y = -x + 2$
(D) $y = x - 1$
(E) $y = -x - 2$
- Q4.** A spacecraft is traveling at a slope of 4. What is the slope of a line perpendicular to its path?
(A) $-\frac{1}{4}$
(B) $\frac{1}{4}$
(C) 4
(D) -4
(E) 0
- Q5.** Two planets are moving in perpendicular paths. If one planet's path has a slope of -2 , what is the slope of the other planet's path?
(A) $\frac{1}{2}$
(B) $-\frac{1}{2}$
(C) 2
(D) -2
(E) 0
- Q6.** What is the equation of a line perpendicular to $y = -3x + 2$ that passes through $(1, 4)$?
(A) $y = \frac{1}{3}x + 3$
(B) $y = -\frac{1}{3}x + 3$
(C) $y = \frac{1}{3}x + 2$
(D) $y = -\frac{1}{3}x + 2$
(E) $y = \frac{1}{3}x + 1$
- Q7.** A scientist is measuring the growth of a plant. If the growth rate is -2 , what is the slope of a line perpendicular to the growth rate?
(A) $\frac{1}{2}$
(B) $-\frac{1}{2}$
(C) 2
(D) -2
(E) 0
- Q8.** What is the equation of a line perpendicular to $y = x + 1$ that passes through $(2, 3)$?
(A) $y = -x + 5$
(B) $y = -x + 4$
(C) $y = -x + 3$
(D) $y = -x + 2$
(E) $y = -x + 1$

Open Ended Questions (FRQ)

- Q9.** A spaceship is traveling through space with a slope of 5. Write the equation of a line perpendicular to its path that passes through the point $(1, 2)$.
- Q10.** Two asteroids are moving through space in perpendicular paths. One asteroid's path has a slope of -4 and passes through the point $(0, 1)$. Write the equation of the other asteroid's path that passes through the point $(2, 3)$.

Chef's Tips

- Use the negative reciprocal to find the slope of a perpendicular line.
- Use the point-slope formula to write the equation of a line.
- Check your work by graphing the lines.